

RESEARCH ARTICLE

Directional coupling of oligodendrocyte connexin-47 and astrocyte connexin-43 gap junctions

Ilaria Fasciani¹ | Paula Pluta² | Daniel González-Nieto³ | Paloma Martínez-Montero⁴ |
Jesús Molano⁴ | Carlos L. Paíno¹  | Oscar Millet² | Luis C. Barrio¹ 

¹Unit of Experimental Neurology and Neurobiology, "Ramón y Cajal" Hospital-IRYCIS, Carretera de Colmenar km 9, Madrid, 28034, Spain

²Structural Biology Unit of CIC bioGUNE, Bizkaia Technology Park, Building 800, Derio, 48160, Spain

³Biomedical Research Networking Center in Bioengineering Biomaterials and Nanomedicine (CIBER-BBN), 28040 Madrid, and Center for Biomedical Technology, Universidad Politécnica de Madrid, Campus de Montegancedo S/N, Pozuelo de Alarcón, Madrid, 28223, Spain

⁴Unit of Molecular Genetics-INGEMM, Hospital "La Paz"-IDIPAZ, Paseo de la Castellana 261, 28046-Madrid, Spain

Correspondence Luis C. Barrio, Unidad de Neurología Experimental, Departamento de Investigación, Hospital "Ramón y Cajal"-IRYCIS. Carretera de Colmenar km 9, 28034-Madrid, Spain.
Email: luis.c.barrio@hrc.es

Funding information

Spanish Government and European Regional Development Fund, Grant/Award Numbers: CSD2008_00005, BUF2012-33821, BUF2015-71078P, CTQ2009-10353; Community of Madrid, Grant/Award Number: S2010/BMD-2460 to L.C.B. and O.M

Abstract

Intercellular communication via gap junction channels between oligodendrocytes and between astrocytes as well as between these cell types is essential to maintain the integrity of myelin in the central nervous system. Oligodendrocyte gap junction connexin-47 (Cx47) is a key element in this crosstalk and indeed, mutations in human Cx47 cause severe myelin disorders. However, the permeation properties of channels of Cx47 alone and in heterotypic combination with astrocyte Cx43 remain unclear. We show here that Cx47 contains three extra residues at 5' amino-terminus that play a critical role in the channel pore structure and account for relative low ionic conductivity, cationic permselectivity and voltage-gating properties of oligodendrocyte-oligodendrocyte Cx47 channels. Regarding oligodendrocyte-astrocyte coupling, heterotypic channels formed by Cx47 with Cx43 exhibit ionic and chemical rectification, which creates a directional diffusion barrier for the movement of ions and larger negatively charged molecules from cells expressing Cx47 to those with Cx43. The restrictive permeability of Cx47 channels and the diffusion barrier of Cx47-Cx43 channels was abolished by a mutation associated with leukodystrophy, the Cx47P90S, suggesting a novel pathogenic mechanism underlying myelin disorders that involves alterations in the panglial permeation.

KEYWORDS

connexin-47, leukodystrophy, oligodendrocyte-astrocyte coupling, pore structure, rectification

1 | INTRODUCTION

Myelin sheath is of vital importance for the "saltatory conduction" of action potentials. A wide variety of devastating myelin disorders arise in the central nervous system as a result of the genetic or autoimmune disruption of key molecules, such as MLC1 and GlialCAM, Kir4.1 potassium channel, CIC-2 chloride channel and AQP4 (Depienne et al., 2013; Knaap, Boor, & Estévez, 2012; Scholl et al., 2009; Wingerchuk, Lennon, Lucchinetti, Pittock, & Weinshenker, 2007), involved in the

siphoning of potassium ions accumulated in myelinated fibers during saltatory conduction, and in the associated and obligatory movements of charge and osmotic water compensation (David, Barrett, & Barrett, 1993; Menichella et al., 2006). In this regard, it has been proposed that the "panglial syncytium" formed by the different types of macroglial cells, including oligodendrocytes (O) and astrocytes (A), connected between them and each other through gap junctions constitutes the principal pathway for the long-distance siphoning of K⁺ ions from the juxtaparanodal periaxonal space to blood vessels or the cerebrospinal fluid (Rash, 2010). Accordingly, it is not surprising that recessive mutations in the *GJC2/GJA12* gene that encodes the oligodendrocyte gap junction protein connexin-47 (Cx47) cause

Ilaria Fasciani and Paula Pluta should be considered joint first author.

Oscar Millet and Luis C. Barrio should be considered joint senior author.

Pelizeaus-Merzbacher-like disease (PMDL1; MIM#608803: Uhlenberg et al., 2004), a prototypic form of hypomyelinating leukodystrophy. Indeed, this disorder is likely caused by mutations that disrupt oligodendrocyte-to-oligodendrocyte (O-O) and oligodendrocyte-to-astrocyte (O-A) gap junctional coupling (Kim, Gloor, & Bai, 2013; Orthmann-Murphy, Enriquez, Abrams, & Scherer, 2007a).

Gap junctions permit the passage of ions and small molecules between cells, thereby providing a basis of direct intercellular communication. Oligodendrocytes and astrocytes express different sets of connexins: oligodendrocytes express Cx47, Cx32 and Cx29 (or its ortholog Cx31.3 in humans) while astrocytes express Cx43 and Cx30, and some of them Cx26 (Altevogt & Paul, 2004; Kamasawa et al., 2005; Nagy, Ionescu, Lynn, & Rash, 2003). Anatomical and functional studies *in vivo* and *in vitro* indicate that O-O coupling is mainly mediated by homotypic Cx47 channels and secondly of Cx32, while heterologous O-A junctions are primarily formed by heterotypic channels of Cx47 with Cx43, but also by Cx32-Cx30 channels depending on the anatomical area (Griemsmann et al., 2015; Maglione et al., 2010; Magnotti, Goodenough, & Paul, 2011; Orthmann-Murphy, Freidin, Fischer, Scherer, & Abrams, 2007b; Tress et al., 2012; Wasseff & Scherer, 2011). Such asymmetric connexin composition of O-A channels may result in a unidirectional instead of a bidirectional flow of ions and larger molecules between coupled cells (Flagg-Newton & Loewenstein, 1980; Suchyna et al., 1999); the presence between oligodendrocytes-astrocytes of a directional diffusion barrier would be relevant to the spatial buffering of periaxonal K^+ as well as for metabolic cooperation and signaling between these cell types. In this regard, it is known that Cx32-Cx30 channels exhibit a marked fast ionic rectification depending on the polarity of transjunctional voltage and that they do not transfer multivalent anions like Lucifer Yellow dye (Orthmann-Murphy et al., 2007b). However, the rectifying nature of Cx47-Cx43 channels remains yet unclear. This issue is hampered by the failure to identify the initiation site for Cx47 translation given the presence in frame of two start codons at the beginning of its protein coding sequence (Met1 and Met4; Figure 1a). Because analysis of the crystal structure of Cx26 channel indicates that amino-terminal domains (NT) are an integral part of the pore structure in which the determinants of permeation reside (Maeda et al., 2009), the composition of Cx47 NT-domain may have functional consequences. Here we established experimentally that the first start site (Met1) is favored over the other (Met4) and that the three extra residues that Cx47 contains at the NT play a critical role in the structure of channel pore, accounting for the restrictive permeation of Cx47 channels and directional coupling of Cx47-Cx43 channels. Finally, we analyzed the effects caused by a mutation associated with leukodystrophy in the permeability of these channels.

2 | MATERIALS AND METHODS

2.1 | Connexin47 constructs and mutagenesis

The coding region of the human Cx47 gene (Genebank NM_020435) containing the two alternative start sites was amplified by PCR from genomic DNA using the forward 5' ACCCGCCCGCCCGCCCTAT 3'

and reverse 5' GAGGCCACCGGCTAAGGAGAAGGCTGA 3' primers. The resulting 1,320 bp fragment (Met1&4Cx47) was cloned into the pBSXG plasmid (González, Gómez-Hernández, & Barrio, 2006) and the bicistronic pIRES2-EGFP plasmid (Clontech Laboratories, Mountain View, CA). A shorter version of 1,311 bp without Met1 and the two following codons (Met4Cx47) was generated by amplifying the plasmids with primers that anneal back-to-back: forward 5'-ATGAGCTGGAGCTTCCTGACGCGGC 3' and reverse 5' GTTGGTCAGAGGGGCGGGCGGGCGGGA 3'. The disease-related P90S substitution was generated by PCR mutagenesis of Met1&4Cx47 construct using the following primers that anneal back-to-back: forward 5' ACCCGCCCGCCCGCCCGCC 3' and reverse 5' ACTAGTG-GAGATGACCACAATCTGGAAGACCC 3'. The human gene encoding the Cx43 coding region (1,500 bp) was obtained by amplifying genomic DNA (Steensel et al., 2005) and it was cloned into pBSXG and pIRES-DsRed plasmids. All constructs were confirmed by DNA sequencing.

2.2 | Expression systems

The preparation of *Xenopus laevis* oocytes, the *in vitro* RNA transcription and microinjection were all performed as described previously (González et al., 2006). Oocytes were injected with an antisense oligonucleotide (15 ng/oocyte) that blocks endogenous Cx38 expression (Barrio et al., 1991), either alone (controls) or with the Met1&4Cx47, Met4Cx47 and Met1&4Cx47P90S or Cx43 RNAs (0.05–0.1 µg/µl, 50 nl/oocyte). After removing the vitelline membrane, injected oocytes were brought into contact in pairs to allow intercellular channels to form. Neuroblastoma cell cultures (Neuro2a; ATCC®, Middlesex, TW, UK) were transiently transfected with pIRES-EGFP/dsRed vectors containing the two forms of Cx47 and the Cx43 DNA inserts as described previously (González et al., 2006).

2.3 | Biotinylation of cell-surface proteins and Western blotting

Oocytes collected 48 hr after RNA injection were incubated for 30 min at 4 °C (35 oocytes per reaction) with 1.5 mg/ml of *N*-hydroxysuccinimide-SS-biotin (ThermoFisher, Rockford, IL), a membrane impermeable biotinylation reagent prepared in ND96 solution (in mM: 96 NaCl, 2 KCl, 1 MgCl₂, 1.8 CaCl₂ and 5 Hepes, [pH 8.0]). After washing five times in ND96 with 100 mM glycine, oocytes were homogenized in lysis buffer (100 mM NaCl, 50 mM Tris-Cl [pH 7.4], 0.5 mM EGTA, 0.5 mM EDTA, 1% Triton X-100, 25 µg/ml leupeptin and 1 mM phenylmethylsulphonylfluoride) in a volume of 10 µl/oocyte and maintained on ice for 30 min. After centrifugation at 16,000g (4 °C) for 15 min, a 50 µl aliquot of the supernatant was mixed 1:1 with SDS-PAGE sample buffer (50 mM Tris-Cl [pH 6.8], 50 mM DTT, 2% SDS, 10% glycerol, 5% βME) to assess the total Cx47 expression (surface plus internal proteins). Then, 50 µl of streptavidin-coated agarose beads (Merck, Kenilworth, NJ, USA) were added to the remaining 300 µl of supernatant, and the samples were left on a roller at 4 °C for 3 hr to allow the streptavidin to bind to the biotinylated proteins. The biotin-streptavidin bead complexes were then washed five times with lysis buffer, recovering the beads after each wash with a 2 min centrifugation (16,000g). The final pellets were dissolved in 30 µl of SDS-PAGE sample buffer and heated at 25 °C for

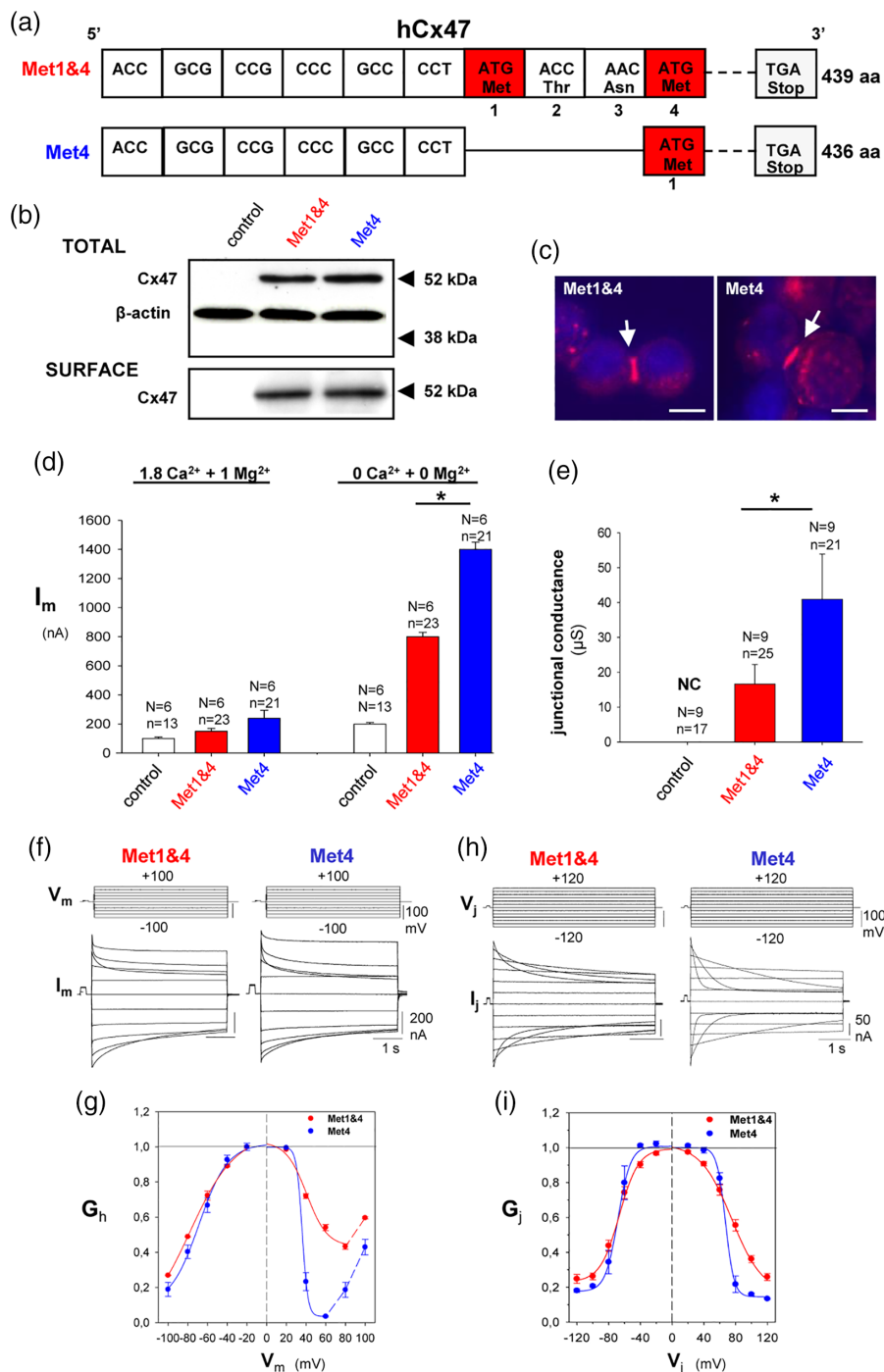


FIGURE 1 Experimental identification of start site for Cx47 synthesis. (a) Two constructs encoding the human Cx47 were generated, one containing the entire coding region with the two alternative Met1 and Met4 start sites (Met1&4) and the other forcing synthesis from Met4 (Met4). (b and c) The total protein and surface expression of Met1&4- and Met4-Cx47 was similar in single oocytes (b), as was the formation of gap junction plaques at the cell contact areas of transfected N2A cells (c; arrows). (d and e) The ionic conductivity of Met4-Cx47 hemichannels (d) and intercellular channels (e) in single and paired oocytes was ~twofold higher than that of Met1&4-Cx47; hemichannel currents activated by depolarizing pulses (from -40 to $+20$ mV, 2 s) in the presence or absence of extracellular divalent cations, and the electrical coupling was measured 24 hr after pairing. (f-i) Voltage regulation of Met1&4- and Met4-Cx47 hemichannels (f and g) and intercellular channels (h and i). Sample recordings from hemichannel currents in single oocytes (I_m) evoked by positive and negative membrane potential pulses (V_m ; ± 100 mV in 20 mV increment and 5 s) applied at $V_h = 0$ mV in the absence of external divalent cations, and those of junctional currents (I_j) between paired oocytes elicited by transjunctional voltage pulses (V_j ; ± 120 mV in 20 mV increment, 5 s) at $V_h = -40$ mV (h). Conductance/voltage curves: Normalized conductances of hemichannel and channel (G_h and G_j) were maximal around 0 mV and decreased progressively as larger negative and positive V_m or V_j gradients were applied. In comparison, hemichannels and channels of Met4-Cx47 were more sensitive to a positive polarity of V_m or to the two polarities of V_j , respectively (see Boltzmann's parameters in Supporting Information Table S1). Differences in ionic conductivity and voltage gating between Met1&4- and Met4-Cx47 based-channels indicate a distinct protein composition: a full-length Cx47 of 439 amino acids and a NT-truncated form of 436 residues. Scale bars, 10 μ m (in b); N, number of experiments; n, number of single or paired oocytes recorded; NC, noncoupled pairs, and * $p < .05$ (in d and e)

2 hr to cleave the disulfide bonds. The proteins were separated in 7.5% SDS-PAGE gels and transferred electrophoretically to nitrocellulose membrane. After blocking with 5% skimmed dry milk in 1% TBS [pH 7.6]/0.1% Tween20 for 1 hr at room temperature, the membranes were incubated overnight at 4 °C with a primary rabbit anti-Cx47 antibody raised against residues 344-399 of the Cx47 carboxyl terminal domain (1:5,000 dilution; kindly provided by J. Orthmann-Murphy). After washing, the primary antibody was detected with a secondary sheep anti-rabbit IgG antibody (1:10,000) conjugated to horseradish peroxidase for 1 hr at room temperature. Antibody binding was visualized by enhanced chemiluminescence (ECL; Amersham Biosciences, Piscataway, NJ) and exposure to X-ray film. To assess molecular weight of blotted proteins Full-range Amersham Rainbow marker (RPN800E) was used, and band intensities were determined by densitometry (ImageQuant TL software V. 2003.02; Amersham Biosciences).

2.4 | Immunocytochemistry

N2a cells collected 24 hr after transfection were fixed for 30 min with 4% paraformaldehyde in 0.1 M PBS [pH 7.4] and permeabilized with ethanol:acetic acid (95% v/v) for 10 min at -20 °C. Cells were preincubated for 30 min in a blocking solution (10% goat serum in PBS) and incubated at 4 °C overnight with the primary Cx47 rabbit antibody (1:1,500). After washing, the cells were incubated at room temperature for 1 hr with a Cy5- or AlexaFluor488-conjugated goat anti-rabbit IgG antibody (1:100; Zymed 81-6116, and 1:500; Invitrogen A11034, ThermoFisher) and mounted (ProlongGold, Invitrogen, Thermo Fisher). Confocal images were acquired at 680/32 or 540/42 nm emission using 647 and 488 nm excitation lines.

2.5 | Dye-uptake and intercellular dye transfer

For dye-uptake experiments, single oocytes were incubated for 30 min at 18 °C with 1 mM propidium iodide (PI; Merck) or 5-(and-6)-carboxyfluorescein (CF; Molecular Probes; Thermo Fisher) dissolved in a ND96 solution ([pH 7.5]) with and without added divalent cations. A group of oocytes were also treated with 500 μ M carbenoxolone from 30 min prior to dye incubation to block hemichannel activity. After washing, confocal images were collected to measure the CF and PI using 488 and 568 nm excitation lines at emission wavelengths of 540/42 and 640/40 nm, respectively ($\times 10$ magnification, MRC-1024 and Lasersharp software; Bio-Rad, Hercules, CA). Two regions of interest (ROI, 15 $\times$ 150 μ m) were defined, one internally at the submembrane area to measure intracellular dye accumulation and another outside the cells for background subtraction. Intercellular dye transfer between homologous and heterologous Cx47-EGFP⁺ and Cx43-DsRed⁺ N2a cell pairs was assessed after intracellular dye injection. The level of expression was estimated from the EGFP/DsRed intensities of the ROI-cell1 (donor) and ROI-cell2 (acceptor: $\sim 12 \mu$ m of diameter per cell, 240 pixels and 255 gray levels). Subsequently, and after reduction of the photodetector gain to avoid interference of reporter proteins with dyes, cell-1 was impaled with a sharp microelectrode filled with a 5% CF

solution in 100 mM K-acetate or 1% PI in 1 M LiCl (250-400 M Ω), and the dyes were iontophoresed respectively with negative and positive current pulses (-0.25/+0.5 nA, 500 ms duration and 1 Hz). The time course of donor cell-1 loading and of the passage of the dye to receptor cell-2 was monitored over 10 and 20 min for CF and PI respectively, and the coupling ratio was calculated as the cell2/cell1 intensity relationship.

2.6 | Electrophysiology

Transmembrane currents (I_m) were recorded in single oocytes using the conventional two-electrode voltage clamp method and junctional currents (I_j) between oocyte pairs with the dual two-electrode voltage-clamp technique (Barrio et al., 1991; González et al., 2006). The microelectrodes filled with (in mM) 3 KCl, 1 EGTA, 5 Hepes [pH 7.2] had a resistance of ~ 0.5 M Ω and the external solution used was ND96 ([pH 7.5]) with and without adding divalent cations. Macroscopic currents were filtered at 200 Hz and sampled at 1 kHz (TL-1 DMA Interface and pClamp 5.5 software; Molecular Devices, Union City, CA).

2.7 | Structure modeling of Cx47 channels

Human Cx26 (entry P29033) and Cx47 (entry Q5T442) protein sequences were obtained from the UniProt database and aligned using the EMBOSS Needle tool from the EBI package (McWilliam et al., 2013). The sequence identity and similarity between Cx26 and Cx47 are respectively 21.4% and 32.4%. Fragments with the lowest homology are found in the cytoplasmic CL and CT domains; the CL loop of Cx47 (His100-Leu209) was found to be highly distorted in the homology model and replaced by the corresponding fragment of a smaller connexin, Cx32, and the CT loop (Ser293-Ile439) was omitted from the model. This substitution did not affect conservative fragments and facilitated later hemichannel construction. Ultimately the sequence identity and similarity between Cx26 and modeled Cx47 are, respectively, 42.3% and 63.6%, considered adequate for homology modeling (Dalton & Jackson, 2007). The full-length and NT-truncated Cx47 structures were obtained by homology modeling based on the high resolution structure of Cx26 (2ZW3) (Maeda et al., 2009). The mutation under consideration was incorporated into the sequence using BioEdit software (Hall, 1999), prior to homology modeling. Different homology servers were tested and the results were cross-validated using the SAVES server (Molecular Biology Institute at the University of California, Los Angeles, CA). The SAVES server employs different algorithms (e.g., PROCHECK and ERRAT's) to check protein structures during and after model refinement: PROCHECK performs a Ramachandran plot analysis. For Cx47, 82.7%, 14.7%, 1.6%, and 1% of residues were in the most favored, additional allowed, generously allowed and disallowed regions, respectively. ERRAT determines the Overall Quality Factor expressed as a percentage of the protein for which the calculated error value falls below the 95% rejection limit, which was 75.691% for Cx47. After validation, the structures with the best scores (according to the CPHmodels-3.2 server results) were further processed (Nielsen, Lundegaard, Lund, & Petersen, 2010). Eventual clashes at this and subsequent stages were



removed manually using the Mutagenesis option of PyMol (The PyMOL Molecular Graphics System, Version 1.5.0.4, Schrödinger, LLC). Any changes to the text files of the structures (renaming chains, renumbering residues, etc.) were performed at the Bioinformatics Software web site. Hemichannels were generated by superposing the target chain on a Cx26 wild type background (2ZW3) using the Magic Fit routine from the program Swiss PDB Viewer (Guex & Peitsch, 1997). Hydrogens were added using the VMD program (Humphrey, Dalke, & Schulten, 1996). All molecular dynamics (MD) simulations were performed with the NAMD2 program (Phillips et al., 2005), along with the CHARMM27 force field (MacKerell et al., 2002). Simulations were performed in the constant volume-constant temperature (NVT) ensemble with the Leapfrog Verlet integrator, employing an integration time step of 2 fs. A protein was solvated in a water sphere and the system was neutralized with NaCl. The central coordinates and the radius of the sphere defined the spherical harmonic boundary conditions with a force constant of 10 kcal·mol⁻¹·Å⁻¹ keeping the system together. To maintain the constant temperature of 300 K the Langevin dynamics with a 5 ps⁻¹ coupling coefficient was applied to nonhydrogen atoms. Covalent bonds between hydrogen atoms and heavy atoms were constrained by means of the RATTLE algorithm. A switching function was used to truncate the nonbonded interactions (Van der Waals and electrostatics) in the cut-off distance of 12 Å. The conjugated gradient method of minimization was used to relax the structure before production through 5,000 force evaluation steps. MD runs were performed for 10 ns (8 ns analyzed) with the co-ordinates written to the trajectory file every 100 steps and the energy values extracted every 10 steps. VMD was used for system preparation and the energy, RMSD and hydrogen bond occupancy was analyzed. The electrostatic surface potential of a hemichannel was calculated using the PDB2PQR Server (Dolinsky, Nielsen, McCammon, & Baker, 2004), and the APBS plugin (Baker, Sept, Joseph, Holst, & McCammon, 2001) for PyMOL under default dielectric constants of 2.0 and 80.0 for the protein and solvent regions, respectively. Molecular representations were created using PyMol and VMD.

2.8 | Statistical analysis

Data (*n*) from replica experiments (*N*) are shown as the mean values ± SEM. Paired and unpaired Student's *t*-test and factorial Analysis of Variance (ANOVA) were used as appropriate (Sigmaplot 11.0, Life Sciences, Erie, PA) and the differences were considered statistically significant at *p* < .05.

3 | RESULTS

3.1 | Permeability and pore structure of Cx47 channels

The exact composition of the NT-domain of Cx47 is currently unknown as the humans *GJC2* gene and that of other vertebrate species contains two start codons at the beginning of coding sequence that are separated by six nucleotides (Met1-X-X-Met4; Figure 1a). Because neither of the two start sites have the A/GXXXATGG motif

that produces efficient translation (Kozak, 1986), the initiation site was determined experimentally. To address this issue, two distinct constructs of human Cx47 were generated: a full-length construct containing the two start sites (Met1&4Cx47) and a second one in which Met1 and the following two codons were deleted to force translation from Met4 (Met4Cx47; Figure 1a). When expressed in *Xenopus laevis* oocytes and N2a cells, these two constructs induced similar levels of Cx47 protein expression and a similar relative abundance at the cell surface, and both proteins forming gap junctional plaques at cell-cell contact areas (Figure 1b,c, and Supporting Information Figure S1A). However, Met1&4 and Met4Cx47 proteins produced hemichannels and intercellular channels with quite different electrophysiological properties. In single oocytes, both types of Cx47 hemichannels were activated upon membrane depolarization in a medium lacking divalent cations and the hemichannel currents recorded in the Met4Cx47 oocytes were ~2-fold greater than those in Met1&4Cx47 oocytes (Figure 1d); these currents were not recorded in control oocytes and were inhibited by 500 μM carbenoxolone (CBX), a known blocker of connexin channels (Supporting Information Figure S2). Also electrical coupling developed by the intercellular channels formed between paired Met4Cx47 oocytes was ~twofold greater than that between Met1&4Cx47 pairs for equal amounts of Cx47 protein expression (Figure 1e and Supporting Information Figure S1B), suggesting that the ionic conductivity of Met4Cx47 hemichannels and intercellular channels is double that of Met1&4Cx47 hemichannels/channels.

In addition, the two types of Cx47 hemichannels and intercellular channels showed distinct voltage gating properties. Both Met1&4 and Met4Cx47 hemichannels exhibited a bipolar dependence on membrane potential (*V_m*), with a maximal macroscopic conductance (*G_h*) around ± 20 mV, and asymmetrical and progressive reductions of *G_h* for the negative and positive polarities of *V_m* as larger gradients were applied (Figure 1f). These properties are consistent with the two voltage-gating mechanisms operating with opposite *V_m* polarities, and referred to in the literature as the "slow-loop" gate and "fast-*V_j*" gate (Fasciani et al., 2013). However, the *G_h*/*V_m* curve for the positive polarity, corresponding to the closure of "fast-*V_j*" gate, had a more pronounced voltage sensitivity in Met4Cx47 hemichannels than in those formed by Met1&4Cx47, whereas the "slow-loop" gate mediating the closing at the negative polarity of *V_m* was indistinguishable in these hemichannels (Figure 1g and Boltzmann's parameters in Supporting Information Table S1). Moreover, Met4Cx47 intercellular channels were also more sensitive to transjunctional voltage (*V_j*) and they had a faster current relaxation (Figure 1h,i and Supporting Information Table S1).

The two classes of Cx47 hemichannels and intercellular channels also exhibited different permselectivity to fluorescent tracers of similar molecular weight and axial dimension, yet with opposite charge: the cationic propidium iodide dye (PI; 414 Da and +2) and the anionic carboxyfluorescein dye (CF; 376 Da and -2). Dye-uptake experiments performed on single Met1&4Cx47 and Met4Cx47 injected oocytes showed poor and strong intracellular dye accumulation over a minor background level 30 min after incubation in the medium in the presence and absence of divalent cations, respectively (Figure 2a,b). This enhanced permeation to dyes in Met1&4Cx47 and Met4Cx47 oocytes was prevented by a pretreatment with 500 μM CBX of

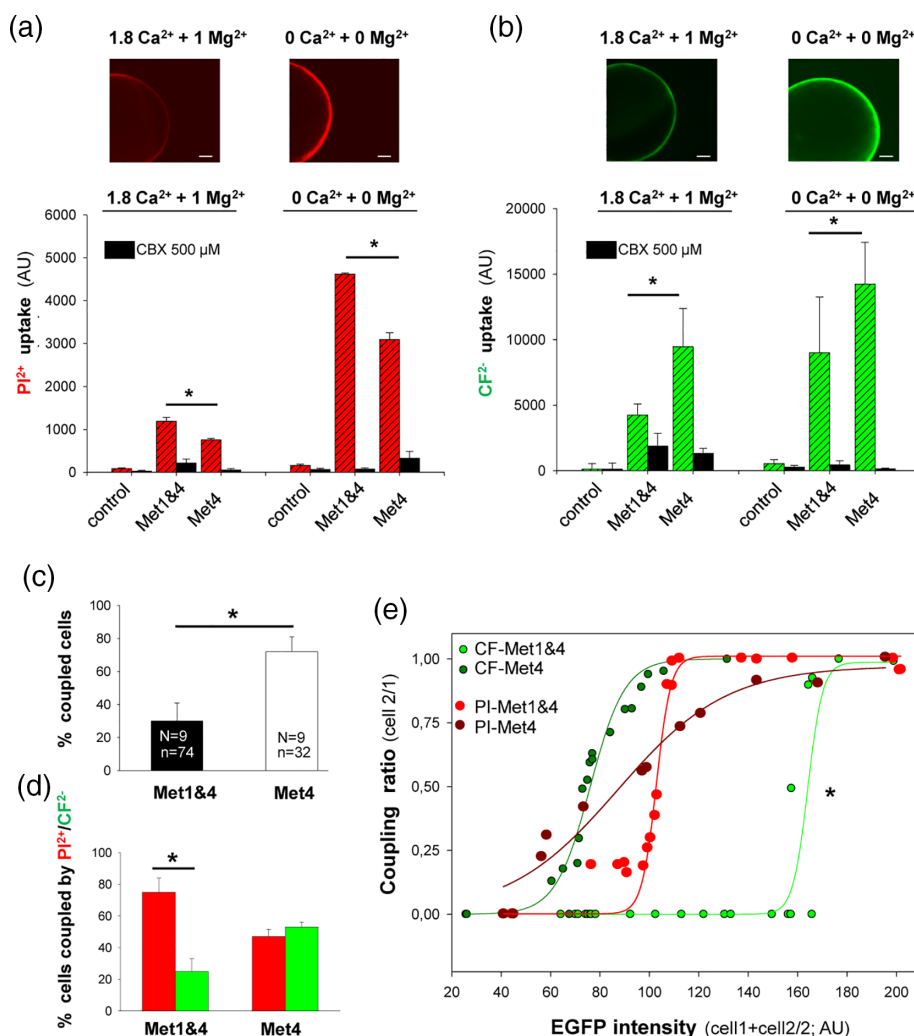


FIGURE 2 Full-length but not NT-truncated Cx47 channels discriminate between cationic and anionic tracers. (a and b) Dye-uptake across Met1&4- and Met4-Cx47 hemichannels. Quantification of the intracellular cationic propidium iodide (PI; in a) and anionic carboxyfluorescein (CF; in b) after a 30 min incubation in the presence or absence of extracellular divalent cations. Dye entry was low under normal conditions and enhanced markedly by lowering the availability of divalent cations, and blocked by 500 μ M carbenoxolone (CBX, see Supporting Information Figure S2); Met1&4-Cx47 hemichannels permeated PI better than CF. (c-e) Intercellular dye-transfer through Met1&4- and Met4-Cx47 channels. Percentage of total dye-coupling and that for each dye, PI and CF (c and d): there was only weak dye-coupling between Met1&4-Cx47 pairs but it was significantly higher for PI than CF while Met4-Cx47 cell pairs coupled more frequently and did not discriminate between cationic and anionic dyes. The coupling ratio was assessed in function of Cx47-EGFP expression (e): transfer of PI and CF between Met4-Cx47 cell pairs, and of PI for Met1&4-Cx47 pairs enhanced similarly when connexin expression increased from low to moderate levels while CF transfer between Met1&4-Cx47 pairs was restricted to very high expression levels. Scale bars, 200 μ m (in a). N, number of experiments; n, total number of cell pairs injected, half with each dye (in c); * $p < .05$ (t-test in A-C; ANOVA in e)

15 min, consistent with a dye entry via hemichannels. PI uptake across Met1&4Cx47 hemichannels was significantly higher than that across Met4-Cx47 hemichannels both in the presence or absence of divalent cations, while CF permeated better through Met4Cx47 than Met1&4Cx47 hemichannels. Studies of intercellular dye transfer performed in N2a cells transiently transfected with Met1&4Cx47- and Met4Cx47-IRES-EGFP plasmids showed that Met1&4Cx47-EGFP⁺ cells were poorly coupled and permeated PI ~3-fold better than CF, while Met4Cx47-EGFP⁺ cells were more frequently coupled and permeated both dyes equally (Figure 2c,d). Dye transfer increased with the level of Cx47 expression, as estimated by the EGFP reporter intensity in the donor and receptor cells; greater transfer of PI and CF occurred in the case of Met4Cx47 pairs and while that of PI in the Met1&4Cx47 pairs was similar, little CF passed between

Met1&4-Cx47 cells and only was detected when expression was particularly strong (Figure 2e; ANOVA $p < .05$). Hence, Met1&4-Cx47 intercellular channels more readily permeate cationic than anionic dyes, while Met4-Cx47 channels do not discriminate between these dyes in function of their charge. These differences in ionic conductivity, charged dye selectivity and voltage gating between the Met1&4Cx47 and Met4Cx47 hemichannels/channels clearly indicate that the proteins translated from the two constructs were distinct: a full length Cx47 of 439 amino acids and another shorter Cx47 version of 436, lacking the first three residues at the 5' extreme of NT-domain of the former (Met₁-Thr₂-Asn₃; Figure 1a).

The structure of full-length (Met1&4) and NT-truncated (Met4) Cx47 channels was assessed on the basis of the only available high resolution crystal structure, that of the Cx26 channel (Maeda et al., 2009),

paying particular attention to the molecular dynamics of the interactions that stabilize the pore structure. Removing the first three NT residues produces important rearrangements affecting the pore and funnel architecture (Figure 3). NT-truncation of the first three residues of Cx47 altered the spatial arrangement of two different groups of interacting residues: Gly42, Tyr47, Arg78, Arg260, Glu263, and Lys264 (Figure 3c, d), and Glu44 and Ser48 (Figure 3e, f); the most strongly affected interaction is the electrostatic interaction between the ^+H_3N -group of Met1 and the carboxylate group of the Glu44 residue of the first α -helix transmembrane domain (TM1) at the narrowest constriction site of the funnel

of pore, which does not occur in the Met4Cx47 channel because the NT-domain is too short (Figure 3a, b). It was also worth noting that the Met1-Met1, Asn3-Asn3, and Asn3-Met4 hydrogen bonds between neighboring connexin chains disappear, which also help to stabilize the funnel (Figure 3b). Thus, NT-truncation appeared to loosen the funnel structure and therefore, the narrower portion of pore widens by $\sim 40\%$ (from $12 \times 10 \text{ \AA}$ to $15 \times 11 \text{ \AA}$) (Figure 3g, h). In addition, this truncation increases the negativity of the electrostatic surface potential profile around the funnel constriction, shifting from a sum of the partial charges of 1.22 to 2.22 (Figure 3i, j).

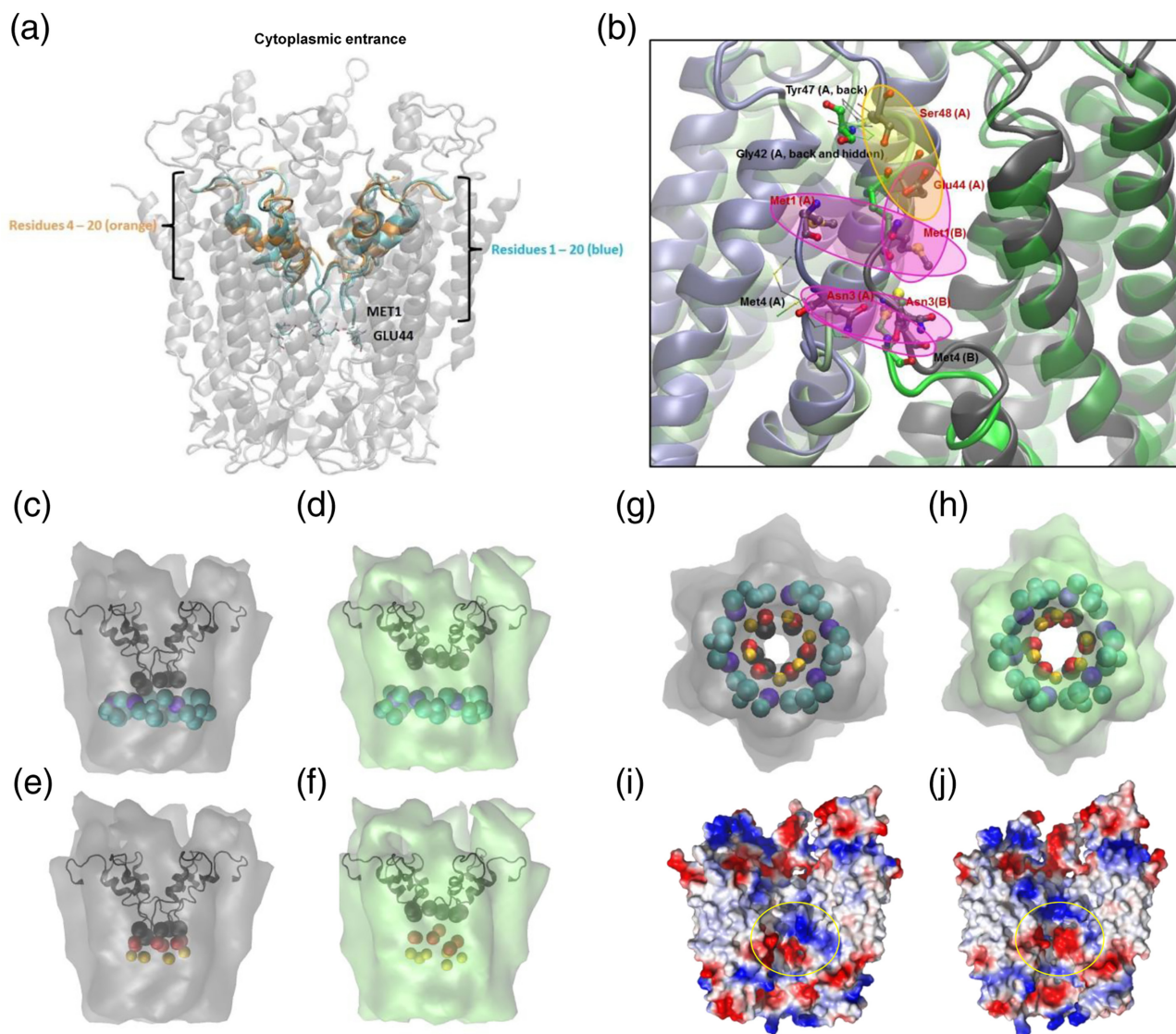


FIGURE 3 Pore structure of Cx47 channel. (a) Vertical cross-section of the full-length and NT-truncated Cx47 hemichannels superimposed (in blue and orange). The NT-domain of full-length Cx47 subunits folds into the pore, lining the inner surface and forming a funnel structure that is strongly stabilized by electrostatic Met1-Glu44 interactions; such interactions do not occur in the NT-truncated domain because it is too short. (b) Side view of full-length NT-domains (light and dark grey) and truncated NT-domains (light and dark green) of two neighboring connexin chains a and b showing interactions linked to the NT-domain affected or lost (yellow and pink ellipses) by removal of the first three N-terminal residues. (c-j) Spatial arrangement of the main residues associated to the full-length (c, e, g, and i) and truncated NT-domains (d, f, h, and j); NT residues 1/4-20 are shown with color beads, Met1 or Met4 (black), Glu44 (red), Ser48 (yellow), Tyr47 (dark blue), and Gly42, Arg78, Arg260, Glu263, and Lys264 (light blue). NT-truncation of the first three residues of Cx47 alters the organization of two different groups of interacting residues: Gly42, Tyr47, Arg78, Arg260, Glu263, and Lys264 (c and d), and Glu44 and Ser48 (e and f); these alterations cause a rearrangement that distorts the funnel structure, increasing the pore diameter (g, h) and the negativity of the electrostatic profile around the pore constriction (i, j; yellow circle). The potentials are displayed with ± 60.1 and 65.1 kTe^{-1} isosurface contours (blue/red) for the full-length and NT-truncated Cx47 channels

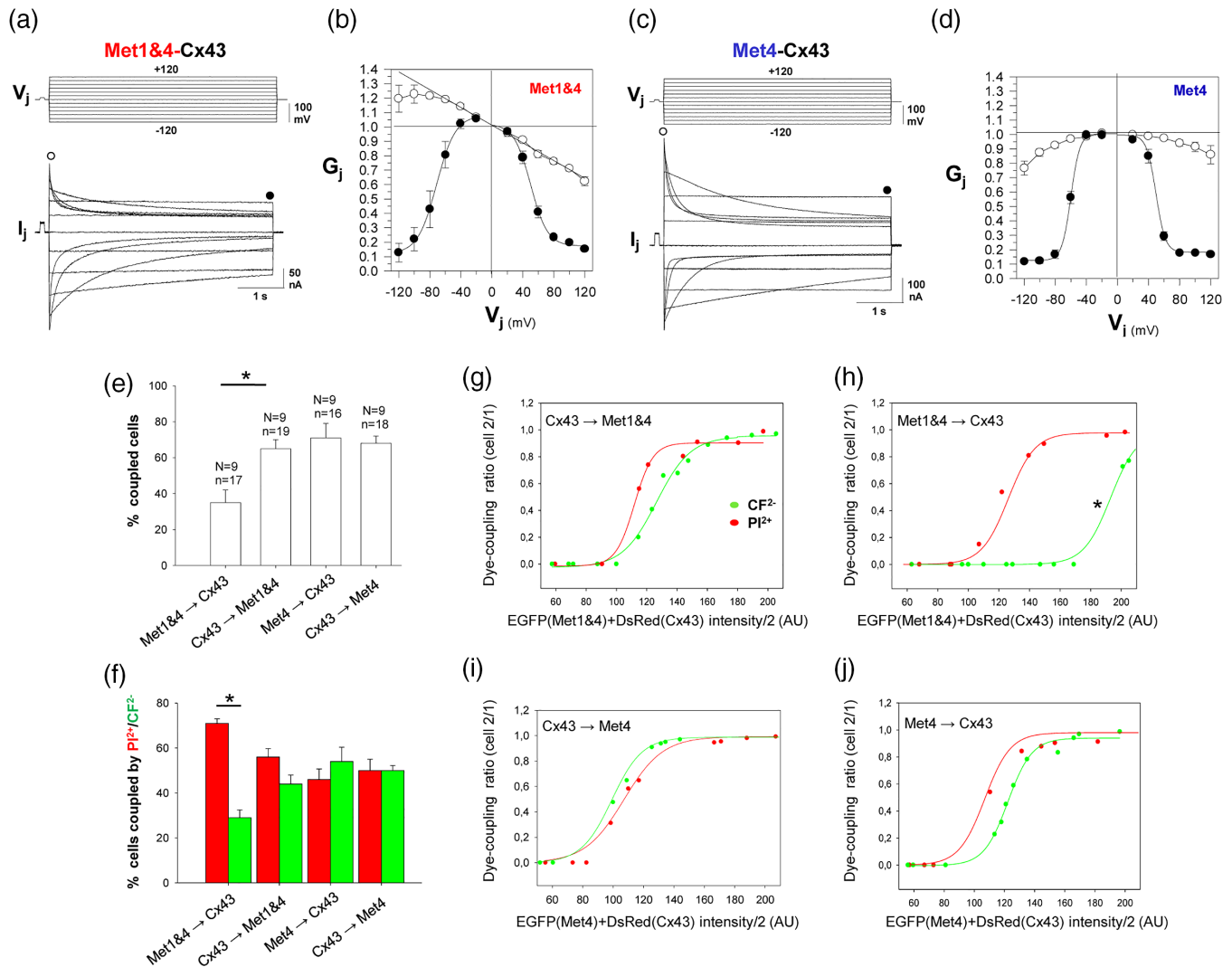


FIGURE 4 Ionic and chemical rectification of oligodendrocyte-astrocyte Cx47-Cx43 channels. (a–d) Fast ionic rectification of heterotypic Met1&4Cx47-Cx43 but not of Met4Cx47-Cx43 channels. The junctional currents (I_j) of Met1&4Cx47-Cx43 oocyte pairs increased progressively less with the larger voltage gradients (V_j) of relative positivity than negativity on the Cx47 cell interior (a); in the case of Met4Cx47/Cx43 pairs, these currents increase symmetrically for the two polarities of V_j (c). Positive and negative V_j 's are defined as relative depolarization and hyperpolarization of Met1&4Cx47 and Met4Cx47 cells, where pulses were applied, and induced respectively upward and downward I_j 's measured in the Cx43 cells. The G_j/V_j relationships show the asymmetric rectification of initial junctional conductance (G_{j0} ; white dots in b and c) of Met1&4Cx47-Cx43 channels with regard to the value at $V_j = 0$, decreasing and increasing with the relative positive and negative V_j on the Cx47 interior while the G_{j0} of the Met4Cx47-Cx43 channels follows an ohmic behavior. In addition, the steady state G_j (G_{jss} ; black dots in b and d) of the two classes of channels also rectified in a time dependent manner for the two polarities of the V_j gradient in accordance with the voltage dependence exhibited by the corresponding homotypic channels (Betzmann's parameters in Supporting Information Table S1). (e–j) Chemical rectification. The directionality of transfer between Met1&4- or Met4Cx47-EGFP⁺ with Cx43-DsRed⁺ N2A cell pairs was studied with an equal number of PI and CF injections into each cell type. Percentage of total dye-coupling and that for each dye, PI and CF (e and f): in the case of Met1&4Cx47-Cx43 pairs, injections into the Cx47 side showed less coupling than those into Cx43 cells and fewer of these cells were coupled for CF than PI, whereas coupling between Met4Cx47-Cx43 pairs was strong and equal in either directions and it was similar for PI and CF. Coupling ratio in function of Cx47-EGFP and Cx43-DsRed expression (g–j): the transfer of PI and CF injected into the Cx43 side toward Met1&4- or Met4-Cx47 cells and that of PI injected into the Met4-Cx47 side to the Cx43 cells increased similarly at the weak and medium expression (g,i,j). However, transfer of CF into Cx43 cells when injected into the Met1&4Cx47 side was restricted to very strong expression (h), revealing a directional diffusion barrier for movement of anionic but not of the cationic dye. N , number of experiments; n , total number of cell pairs injected, half with each dye (in c); * $p < .05$ (t-test in e,f, and ANOVA in h)

3.2 | Directional ionic flux and dye transfer through oligodendrocyte Cx47-astrocyte Cx43 channels

The full-length and NT-truncated Cx47 formed heterotypic channels with astrocyte Cx43 that had distinct electrical coupling ($19.1 \pm 7.0 \mu S$ vs. $41.3 \pm 5.2 \mu S$; $p < .05$), and asymmetric and symmetric properties of ionic conductivity and dye transfer, respectively (Figure 4). In contrast to

the ohmic behavior of the NT-truncated Cx47-Cx43 channels, the initial junctional conductance of full-length Cx47-Cx43 channels rectified fast and asymmetrically with the polarity of transjunctional voltage, increasing and decreasing relative to its value at $V_j = 0$ for relatively negative and positive V_j 's on the Cx47 cell's interior ($\sim$ twofold between ± 100 mV; Figure 4a–d). Dye transfer across heterotypic channels of full-length but

not NT-truncated Cx47 with Cx43 was also asymmetrical (Figure 4e–j). Following dye injection into full-length Cx47-EGFP⁺ cells, there was little dye coupling with Cx43DsRed⁺ cells, although it was significantly higher for PI than CF (Figure 4e,f). Conversely, there was more coupling evident when the dye was injected into the Cx43DsRed⁺ cells, and it was similar for the anionic and cationic dyes. In contrast, the dye coupling between heterotypic NT-truncated Cx47-EGFP⁺-Cx43-DsRed⁺ cell pairs was stronger and similar for both dyes, irrespective of which cell was injected. In accordance with this behavior, the increment of transfer of CF but not PI as expression increased was asymmetrical in the case of full-length Cx47/Cx43 cell pairs: transfer increased similarly for CF and PI in injections into the Cx43 side, or of PI into the Cx47 cell, yet transfer was restricted even at the highest levels of expression when CF was injected into the Cx47 side (Figure 4g,h; ANOVA $p < .05$). The increment of transfer of the two dyes was however similar in both directions across the heterologous NT-truncated Cx47-EGFP⁺-Cx43-DsRed⁺ cell pairs (Figure 4i,j).

3.3 | A mutation of leukodystrophy that alters the permeability properties of Cx47-Cx47 and Cx47-Cx43 channels

Most recessive mutations in Cx47 that cause hypomyelinating leukodystrophy disrupt the capacity of this connexin to form homotypic Cx47-Cx47 and heterotypic Cx47-Cx43 channels (Kim et al., 2013; Orthmann-Murphy et al., 2007a). However, here we reevaluated one of these PMLD1 mutations, the P90S substitution in the second α -helix transmembrane domain (TM2) (Uhlenberg et al., 2004), which seems to retain partially the ability to form functional intercellular channels (Orthmann-Murphy et al., 2007a). Oocytes expressing the full-length Cx47P90S (P90S) produced a similar total amount of this Cx47 protein expression as those expressing the WT, yet this Cx47P90S was less abundant at the cell surface (Figure 5a,b) and smaller hemichannel currents were evident upon depolarization in the absence of extracellular divalent cations (Figure 5c). N2a cells transfected with the P90S-IRES-EGFP plasmid formed gap junctional plaques in cell-to-cell contact areas but fewer than those found between WT-IRES-DsRed transfected cells (Figure 5d). The dye coupling among P90S-EGFP⁺ cells was also markedly weaker but, in contrast to WT channels, the P90S channels did not discriminate between anionic and cationic dyes (Figure 5e,f). Homotypic P90S oocyte pairs coupled electrically, albeit significantly less than WT pairs (Figure 5g). The voltage-gating properties of homotypic P90S-P90S channels were undistinguishable to those of WT-WT channels (Figure 5h,i; and Supporting Information Table S1), yet heterotypic P90S-WT channels exhibited fast rectification of the initial G_j , increasing and decreasing progressively with a relatively larger negative and positive V_j on the P90S cell interior (Figure 5j,k; Supporting Information Table S1). When the Cx47P90S mutant was combined with astrocyte Cx43, similar high levels of electrical coupling were induced as in the WT-Cx43 pairs (Figure 5g). However, the P90S/Cx43 channels did not display the characteristic fast ionic rectification with V_j seen in wild type full-length Cx47-Cx43 channels (Figure 5l,m; Supporting Information Table S1).

4 | DISCUSSION

The dilemma produced by the two alternative start sites for human Cx47 synthesis (Met1-X-X-Met4) has been resolved experimentally by combining mutagenesis and electrophysiology. Thus, the first conclusion drawn from this study is that the translation of Cx47 starts with the linear screening of the mRNA at the first available methionine (Met1), in agreement with the prediction obtained by several algorithms for proteins with a Met1 in a “weak” Kozak’s context (<http://bioinfo.iitk.ac.in/AUGPred/>). The start site does not affect translation efficiency, protein processing or hemichannel forming ability, as was previously reported (Diekmann, Henneke, Burckhardt, & Gärtner, 2010), however depending on start site hemichannels/channels exhibited marked differences in ionic conductivity, permeability and voltage-gating properties. Some of biophysical properties show here for human NT-truncated (Met4) Cx47 channels had been previously reported, such are its stronger V_j dependence and no selectivity on the bases of dye charge (Orthmann-Murphy et al., 2007b; Teubner et al., 2001), as well as the weaker V_j sensitivity observed for full-length M1&4-Cx47 channels with GFP tagged at the carboxyl terminus (Kim et al., 2013). However, other differential properties of full-length Cx47 channels like its lower ionic conductivity and the cationic over anionic dye selectivity had not yet been identified. As previously reported, the full-length and truncated Cx47 combine efficiently with itself and with astrocyte Cx43 (Kim et al., 2013; Orthmann-Murphy et al., 2007a) while they do not form functional channels with oligodendrocyte Cx32 (Supporting Information Figure S3A; Magnotti et al., 2011; Orthmann-Murphy et al., 2007a) and astrocyte Cx26 (Supporting Information Figure S4; Magnotti et al., 2011). Also, we found that full-length Cx47 can form heterotypic channels of rectifier type with oligodendrocyte Cx31.3 (Supporting Information Figure S3B), and astrocyte Cx30 (Supporting Information Figure S4A), which is consistent with co-localization of Cx47 and Cx30 *in vivo* and heterotypic dye coupling reported *in vitro* (Altevogt & Paul, 2004; Magnotti et al., 2011), but the level of electrical coupling detected in these two heterotypic combinations is too small to be functionally relevant.

Comparing the biophysical properties of the full-length and NT-truncated Cx47 hemichannels/channels clearly indicates that the first three residues of the NT-domain (Met₁-Thr₂-Asn₃) constitutes an integral part of the permeation pathway that accounts for the relatively low ionic conductivity and the cationic over anionic dye selectivity of full-length Cx47 channels. NT-domains of six connexin subunits fold into the channel pore, lining the inner surface and causing the pore to gradually reduce its diameter from the cytoplasmic surface to the midpoint of plasma membrane, establishing a funnel structure (Maeda et al., 2009). In this regard, molecular dynamics simulations of Cx47 channels reveal that of all the intersubunit and intrasubunit hydrogen bonds, and the electrostatic interactions linked to the NT-domain, the strong electrostatic interaction between the Met1 and Glu44 residue of TM1 at the inner portion of the pore plays a pivotal role in the maintenance of funnel architecture and its regulation. This Met1(NT)-Glu44(TM1) interaction is unique to full-length Cx47 because NT-domains of other connexins like Met4Cx47 are too

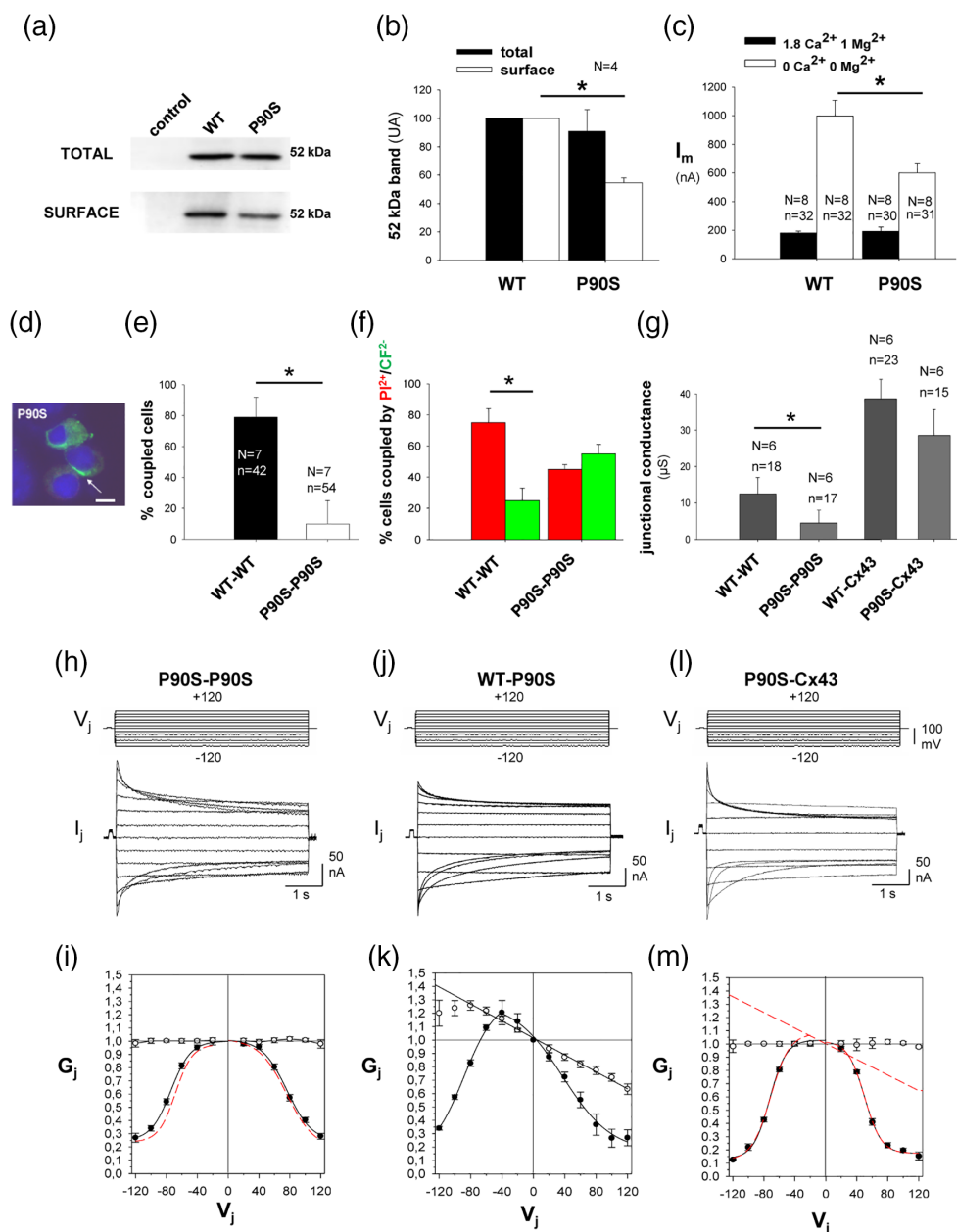


FIGURE 5 Cx47P90S mutation of leukodystrophy abolishes the permselectivity of Cx47-Cx47 and Cx47-Cx43 channels. (a-c) Hemichannel function. The total protein expression of Cx47P90S (P90S) in injected oocytes was similar to that of the full-length wild type Cx47 (WT) but it was less abundant at the cell surface (a and b); P90S also produced smaller hemichannel currents activated upon +20 mV depolarization in the absence of extracellular divalent cations (c). (d-g) Intercellular channel function. P90S in transfected N2a cell pairs was able to form gap junctional plaques (d, arrow) but it induced weaker dye-coupling than WT, and channels transferred the anionic CF and cationic PI equally, in contrast to the cationic preference over the anionic dye of WT channels (e and f). The homotypic P90S oocyte pairs were more weakly coupled electrically than the WT pairs, while in heterotypic combination with Cx43 oocyte pairs coupled similarly to WT-Cx43 pairs (g). (h-m) Novel rectifying properties of P90S-based intercellular channels. The V_j dependence of homotypic P90S-P90S channels was indistinguishable from that of WT-WT channels (h and i, black and red curves) while heterotypic P90S-WT channels exhibited a well-pronounced fast asymmetrical ionic rectification with V_j polarity (j and k, white dots); heterotypic P90S-Cx43 channels did not display the fast rectification characteristic of WT-Cx43 channels (l and m, black and red lines; Boltzmann's parameters in Supporting Information Table S1). Positive and negative polarities of V_j are defined as relative depolarization and hyperpolarization of WT (in j) and P90S (in l) cells, where the pulses were applied, and induced upward and downward I_j 's recorded respectively in P90S and Cx43 cells. Scale bars, 25 μm (in d); N, number of experiments; n, number of single or paired cells recorded or injected, half with each dye, and *, $p < .05$ (in b,c and e-g)

short (Maeda et al., 2009). The spatial rearrangements provoked by removing the first three residues of the NT-domain of Cx47 distort the funnel structure. Accordingly, the pore diameter widens and the negativity of the electrostatic potential around the pore constriction site enhances, increasing ionic conductivity and anionic dye

permeability. Charged residues located in an equivalent position of Glu44 in other connexins (e.g., the Glu41 in Cx32 or Glu42Asp43 in Cx46) have been reported to be important determinants of charge selectivity filter (Oh, Verselis, & Bargiello, 2008; Trexler, Bukauskas, Kronengold, Bargiello, & Verselis, 2000). NT-truncation also increases



the voltage sensitivity of “fast- V_j ” gating of hemichannels and channels without affecting the mechanism of “slow-loop” gating. Based on the structure of the Cx26 channel in the open and closed states (Maeda et al., 2009; Oshima, Tani, Hiroaki, Fujiyoshi, & Sosinsky, 2007), it has been proposed that in the absence of a voltage gradient across the lumen pore (i.e., $V_j = 0$), the “fast- V_j ” gate is in the open state when the six NT-helices forming the pore funnel are attached to the pore wall formed by the TM1 domains. If the electrical potential inside the pore is positive, the six α -helices in which the voltage sensor is thought to reside release from the wall and form a plug structure, which partially occludes the permeation pathway. Taking into account that none of the charged residues considered to be part of the voltage sensor are located in the NT-truncation region of Cx47, the apparent increase in voltage sensitivity is more likely to be ascribed to the reduced energy barrier between the open to closed states due to the disappearance of strong Met1(NT)-Glu44(TM1) interactions, and of multiple intersubunit and intrasubunit hydrogen bonds that stabilize the funnel structure of the full-length Cx47 channel.

Another original finding of present study is that heterotypic channels formed by full-length Cx47 with astrocyte Cx43 create a directional barrier to the diffusion of negatively charged molecules. Ionic conductivity through full-length Cx47-Cx43 channels exhibits a well-pronounced fast asymmetrical rectification, augmenting and diminishing respectively with the relative negativity and positivity of V_j on the Cx47 side. However, those channels formed between NT-truncated Cx47 with Cx43 do not rectify as was previously reported for mouse Met4-Cx47/Cx43 channels (Orthmann-Murphy et al., 2007b). This kind of “time-independent” ionic rectification was initially described in Cx26-Cx32 channels (Barrio et al., 1991) and later in many other heterotypic connexin junctions. Such rectification can be fully explained at the single channel level by the rectifying behavior of the main open conductance (Bukauskas, Elfgang, Willecke, & Weingart, 1995). Mechanistically, this rectification derives from the in series composition of the two hemichannels with different ionic conductance and selectivity; this arrangement produces a voltage polarity dependent accumulation or depletion of ions within the channel pore, and therefore, conductance is high or low, respectively (Suchyna et al., 1999). The fast ionic rectification of heterotypic Met1&4Cx47-Met4Cx47 channels is also consistent with these postulates (Supporting Information Figure S5). Under a chemical gradient, heterotypic full-length Cx47-Cx43 channels also display asymmetrical permeability to anionic CF^{2-} while cationic PI^{2+} dye passes in either direction equally well. Deviation of permeability symmetry for large molecules has been previously reported in other heterotypic combinations (Flagg-Newton & Loewenstein, 1980; Robinson, Hampson, Munro, & Vaney, 1993). All of them share the same principles: i) a channel more restrictive on one side than on the other since it is built by two hemichannels with different permselectivity; CF^{2-} dye passes readily through Cx43 channels which does not discriminate significantly between anionic and cationic molecules (Supporting Information Figure S6; Kanaporis, Brink, & Valiunas, 2011; Wang & Veenstra, 1997), while it passes only marginally across more restrictive full-length Cx47 channels, and ii) a favored direction that is the one where the selective permeant first passes through less restrictive hemichannel (i.e., Cx43). To this

respect, our results indicate that funnel structure of full-length Cx47 hemichannel is essential for generating asymmetrical permeability because it has a strong detrimental effect on the flux of anionic large molecules, while directional selectivity could be reasonably derived of in series apposition of Cx43 hemichannel with a positive atmosphere around the intracellular channel entrance, which would elevate concentration of negatively charged molecules within pore and hence increases the flux (Maeda et al., 2009; Weber, Chang, Spaeth, Nitsche, & Nicholson, 2004).

Because Met1&4 and Met4 forms of Cx47 were here heterologously expressed, it needs yet to be confirmed in future studies if oligodendrocytes also use the first start codon to translate Cx47. However, the poor permeability to anionic dyes observed here for homotypic full-length Cx47 channels correlates well with the weak O-O coupling in the optic nerve when assessed with biocytin (372 Da, uncharged) and virtual uncoupling for Lucifer Yellow (443 Da, -2 charge; Robinson et al., 1993), and no coupling for Lucifer Yellow or Alexa fluor 594 (819 Da, -1 charge) and the more extensive coupling for biocytin observed in mouse slices of corpus callosum and cerebellar white matter (Berger, Schnitzer, & Kettenmann, 1991; Magnotti et al., 2011; Wasseff & Scherer, 2011). The presence of a directional diffusion barrier between oligodendrocytes and astrocytes was initially proposed two decades ago based on the fact that Lucifer Yellow dyes injected into astrocytes of the rabbit optic nerve always pass to oligodendrocytes but not in the opposite direction (Robinson et al., 1993); it is consistent with the preferential transfer of anionic CF from cells expressing Cx43 to the interior of Cx47 cells reported here. O-A coupling in the optic nerve, spinal cord, corpus callosum and neocortex is dependent of Cx47-Cx43 channels (Maglione et al., 2010; Magnotti et al., 2011; Menichella, et al., 2006). In the thalamus with a significant proportion of astrocytes lacking Cx43 but abundantly expressing Cx30 (Griemsmann et al., 2015), O-A coupling is likely mediated by Cx32-Cx30 channels, which also would create a diffusion barrier since these channels rectify instantaneously in the same direction that Cx47-Cx43 channels do and they do not transfer Lucifer Yellow (Supporting Information Figure S4B; Orthmann-Murphy et al., 2007b). Accordingly, networks of oligodendrocytes and astrocytes may represent two intercellular compartments with different permeation properties that are coupled through a directional diffusion barrier which limits the transfer of negatively charged molecules in a size-dependent manner from oligodendrocytes to astrocytes but not in the opposite direction. O-A coupling may be implicated in the spatial buffering of K^+ ions accumulated in myelinated fibers during saltatory conduction that may be largely accomplished by a long distance mechanism of K^+ siphoning; a process driven by an electrochemical gradient in which K^+ firstly enters oligodendrocytes, and then passes into the network of coupled astrocytes through heterogenous junctions of Cx47-Cx43 or Cx32-Cx30 channels. The ionic rectification of Cx47-Cx43 channels observed in our conditions of symmetrical intracellular K^+ predicts that the efflux of K^+ and influx of Cl^- evoked by positive transjunctional voltages into Cx47 cell interior relative to Cx43 side is enhanced less potently than expected for a linear nonrectifying conductance. Hence, the detrimental effect on the movement of K^+ ions from oligodendrocytes to astrocytes would initially force a redistribution of K^+ within the network of coupled oligodendrocytes,

favoring the local recycling of K^+ ions. However, a higher increase in K^+ in oligodendrocytes might alleviate the ionic rectification (Suchyna et al., 1999), aiding the efflux of K^+ ions toward astrocytes. It is likely that intercellular A-O communication also constitutes a route for the transfer of metabolites and other signals that oligodendrocytes require for myelin synthesis or to maintain the integrity of the myelin and axon (Hirrlinger & Nave, 2014). In this context, chemical rectification of the O-A barrier may facilitate an efficient supply of anionic metabolites and second messengers to oligodendrocytes by astrocytes, such as pyruvate, lactate, ATP, cAMP or IP_3 , while their flux in the opposite direction would be less favored.

There are several lines of evidence indicating that PMDL1 type leukodystrophy is caused by recessive mutations in Cx47 that prevents them from forming homotypic channels and heterotypic channels with Cx43 (Kim et al., 2013; Orthmann-Murphy et al., 2007a); as such, the vacuolization of myelin associated with this condition may result from disrupted O-O and O-A intercellular communication, and from the subsequent increase in periaxonal K^+ . This phenomenon would lead to a blockage of axonal conduction by depolarization, water retention and finally myelin destruction. We report here a new mechanism underlying the gain-of-function associated with the disease-related Cx47P90S mutation (Uhlenberg et al., 2004), which does not disrupt completely the capacity to form functional homotypic and heterotypic channel with astrocyte Cx43 but abolishes the restrictive permeability of the Cx47 channels formed as well as the ionic and chemical rectification of Cx47-Cx43 channels. Structurally, this proline is conserved in the middle of TM2 of all connexins and it perturbs the α -helix by introducing a kink that bends the cytoplasmic portion of TM2, displacing TM1 toward the luminal side of the pore (Maeda et al., 2009). Thus, suppressing this TM2 proline-kink by introducing the Cx47P90S mutation could straighten TM2 helix, altering the pore geometry to the extent that the charge selectivity filter disappears. This striking result suggests that preserving the restrictive permeability of O-O coupling, the presence of O-A diffusion barrier or both might play a critical role in the way that glia support myelin function and that impairing these phenomena might constitute a novel pathogenic mechanism for myelin disorders. To this regards, we hypothesize that mutation may cause the leukodystrophy by altering electrochemical gradient, making K^+ siphoning less efficient, and/or by affecting to metabolic cooperation and signaling between glial cells.

ACKNOWLEDGMENTS

This work was supported by grants from the Spanish Government and European Regional Development Fund (CSD2008_00005, BUF2012-33821, BUF2015-71078P and CTQ2009-10353) and the Community of Madrid (S2010/BMD-2460) to L.C.B. and O.M. We express our gratitude to J. Orthmann-Murphy (Pennsylvania University) for kindly providing the anti-Cx47 antibodies, and Amparo Martin for her technical assistance.

CONFLICT OF INTEREST

None of the authors have any conflict of interest.

ORCID

Carlos L. Paño  <https://orcid.org/0000-0003-4245-592X>

Luis C. Barrio  <https://orcid.org/0000-0002-4948-4371>

REFERENCES

- Altevogt, B. M., & Paul, D. L. (2004). Four classes of intercellular channels between glial cells in the CNS. *The Journal of Neuroscience*, 24(18), 4313–4323.
- Baker, N. A., Sept, D., Joseph, S., Holst, M. J., & McCammon, J. A. (2001). Electrostatics of nanosystems: Application to microtubules and the ribosome. *Proceeding of the National Academy of Sciences United States of America*, 98(18), 10037–10041.
- Barrio, L. C., Suchyna, T., Bargiello, T., Xu, L. X., Roginski, R. S., Bennett, M. V., & Nicholson, B. J. (1991). Gap junctions formed by connexins 26 and 32 alone and in combination are differently affected by applied voltage. *Proceeding of the National Academy of Sciences United States of America*, 88(19), 8, 8410–8414.
- Berger, T., Schnitzer, J., & Kettenmann, H. (1991). Developmental changes in the membrane current pattern, K^+ buffer capacity, and morphology of glial cells in the corpus callosum slice. *The Journal of Neuroscience*, 11(10), 3008–3024.
- Bukauskas, F. F., Elfgang, C., Willecke, K., & Weingart, R. (1995). Heterotypic gap junction channels (connexin26–connexin32) violate the paradigm of unitary conductance. *Pflügers Archiv: European Journal of Physiology*, 429(6), 870–872.
- Dalton, J. A., & Jackson, R. M. (2007). An evaluation of automated homology modelling methods at low target template sequence similarity. *Bioinformatics*, 23(15), 1901–1908.
- David, G., Barrett, J. N., & Barrett, E. F. (1993). Activation of internodal potassium conductance in rat myelinated axons. *The Journal of Physiology*, 472, 177–202.
- Depienne, C., Bugiani, M., Dupuits, C., Galanaud, D., Touitou, V., Postma, N., ... van der Knaap, M. S. (2013). Brain white matter oedema due to CIC-2 chloride channel deficiency: An observational analytical study. *Lancet Neurology*, 12(7), 659–668.
- Diekmann, S., Henneke, M., Burckhardt, B. C., & Gärtner, J. (2010). Pelizaeus-Merzbacher-like disease is caused not only by a loss of connexin47 function but also by a hemichannel dysfunction. *European Journal of Human Genetics*, 18(9), 985–992.
- Dolinsky, T. J., Nielsen, J. E., McCammon, J. A., & Baker, N. A. (2004). PDB2PQR: An automated pipeline for the setup of Poisson–Boltzmann electrostatics calculations. *Nucleic Acids Research*, 32(Web Server), W665–W667.
- Fasciani, I., Temperán, A., Pérez-Atencio, L. F., Escudero, A., Martínez-Montero, P., Molano, J., ... Barrio, L. C. (2013). Regulation of connexin hemichannel activity by membrane potential and the extracellular calcium in health and disease. *Neuropharmacology*, 75, 479–490.
- Flagg-Newton, J. L., & Loewenstein, W. R. (1980). Asymmetrically permeable membrane channels in cell junction. *Science (New York, N.Y.)*, 207(4432), 771–773.
- González, D., Gómez-Hernández, J. M., & Barrio, L. C. (2006). Species specificity of mammalian connexin26 to form open voltage-gated hemichannels. *FASEB Journal*, 20(13), 2329–2338.
- Griemsmann, S., Höft, S. P., Bedner, P., Zhang, J., von Staden, E., Beinbauer, A., ... Steinhäuser, C. (2015). Characterization of panglial gap junction networks in the thalamus, neocortex, and hippocampus reveals a unique population of glial cells. *Cerebral Cortex*, 25(10), 3420–3433.
- Guex, N., & Peitsch, M. C. (1997). SWISS-MODEL and the Swiss-PdbViewer: An environment for comparative protein modeling. *Electrophoresis*, 18(15), 2714–2723.
- Hall, T. (1999). BioEdit: A user-friendly biological sequence alignment editor and analysis program for Windows 95/98/NT. *Nucleic Acids Symposium Series*, 41, 95–98.
- Hirrlinger, J., & Nave, K. A. (2014). Adapting brain metabolism to myelination and long-range signal transduction. *Glia*, 62(11), 1749–1761.
- Humphrey, W., Dalke, A., & Schulten, K. (1996). VMD: Visual molecular dynamics. *Journal of Molecular Graphics*, 14(1), 33–38. 27–28.



- Kamasawa, N., Sik, A., Morita, M., Yasumura, T., Davidson, K. G., Nagy, J. I., & Rash, J. E. (2005). Connexin-47 and connexin-32 in gap junctions of oligodendrocyte somata, myelin sheaths, paranodal loops and Schmidt-Lanterman incisures: Implications for ionic homeostasis and potassium siphoning. *Neuroscience*, 136(1), 65–86.
- Kanaporis, G., Brink, P. R., & Valiunas, V. (2011). Gap junction permeability: Selectivity for anionic and cationic probes. *American Journal of Physiology-Cell Physiology*, 300(3), C600–C609.
- Kim, M. S., Gloor, G. B., & Bai, D. (2013). The distribution and functional properties of Pelizaeus-Merzbacher-like disease-linked Cx47 mutations on Cx47/Cx47 homotypic and Cx47/Cx43 heterotypic gap junctions. *Biochemical Journal*, 452(2), 249–258.
- Knaap, M. S., Boor, I., & Estévez, R. (2012). Megalencephalic leukoencephalopathy with subcortical cysts: Chronic white matter oedema due to a defect in brain ion and water homeostasis. *Lancet Neurology*, 11(11), 973–985.
- Kozak, M. (1986). Point mutations define a sequence flanking the AUG initiator codon that modulates translation by eukaryotic ribosomes. *Cell*, 44(2), 283–292.
- MacKerell, A. D., Brooks, B., Brooks, C. L., Nilsson, L., Roux, B., Won, Y., & Karplus, M. (2002). CHARMM: The Energy Function and Its Parameterization. In P. R. Schreiner (Ed.), *Encyclopedia of computational chemistry* (Vol. 1, pp. 271–277). John Wiley & Sons.
- Maeda, S., Nakagawa, S., Suga, M., Yamashita, E., Oshima, A., Fujiyoshi, Y., & Tsukihara, T. (2009). Structure of the connexin 26 gap junction channel at 3.5 Å resolution. *Nature*, 458(7238), 597–602.
- Maglione, M., Tress, O., Haas, B., Karram, K., Trotter, J., Willecke, K., & Kettenmann, H. (2010). Oligodendrocytes in mouse corpus callosum are coupled via gap junction channels formed by connexin47 and connexin32. *Glia*, 58(9), 1104–1117.
- Magnotti, L. M., Goodenough, D. A., & Paul, D. L. (2011). Functional heterotypic interactions between astrocyte and oligodendrocyte connexins. *Glia*, 59(1), 26–34.
- Menichella, D. M., Majdan, M., Awatramani, R., Goodenough, D. A., Sirkowski, E., Scherer, S. S., & Paul, D. L. (2006). Genetic and physiological evidence that oligodendrocyte gap junctions contribute to spatial buffering of potassium released during neuronal activity. *Journal Neuroscience*, 26(43), 10984–10991.
- McWilliam, H., Li, W., Uludag, M., Squizzato, S., Park, Y. M., Buso, N., ... López, R. (2013). Analysis tool web services from the EMBL-EBI. *Nucleic Acids Research*, 41(Web Server issue), W597–W600.
- Nagy, J. I., Ionescu, A. V., Lynn, B. D., & Rash, J. E. (2003). Coupling of astrocyte connexins Cx26, Cx30, Cx43 to oligodendrocyte Cx29, Cx32, Cx47: Implications from normal and connexin32 knockout mice. *Glia*, 44(3), 205–218.
- Nielsen, M., Lundegaard, C., Lund, O., & Petersen, T. N. (2010). CPHmodels-3.0-remote homology modeling using structure-guided sequence profiles. *Nucleic Acids Research*, 38(suppl_2), W576–W581.
- Oh, S., Verselis, V. K., & Bargiello, T. A. (2008). Charges dispersed over the permeation pathway determine the charge selectivity and conductance of a Cx32 chimeric hemichannel. *The Journal of Physiology*, 586(10), 2445–2461.
- Orthmann-Murphy, J. L., Enriquez, A. D., Abrams, C. K., & Scherer, S. S. (2007a). Loss-of-function GJA12/Connexin47 mutations cause Pelizaeus-Merzbacher-like disease. *Molecular and Cellular Neuroscience*, 34(4), 629–641.
- Orthmann-Murphy, J. L., Freidin, M., Fischer, E., Scherer, S. S., & Abrams, C. K. (2007b). Two distinct heterotypic channels mediate gap junction coupling between astrocyte and oligodendrocyte connexins. *Journal Neuroscience*, 27(51), 13949–13957.
- Oshima, A., Tani, K., Hiroaki, Y., Fujiyoshi, Y., & Sosinsky, G. E. (2007). Three-dimensional structure of a human connexin26 gap junction channel reveals a plug in the vestibule. *Proceeding of the National Academy of Sciences United States of America*, 104(24), 10034–10039.
- Phillips, J. C., Braun, R., Wang, W., Gumbart, J., Tajkhorshid, E., Villa, E., ... Schulten, K. (2005). Scalable molecular dynamics with NAMD. *Journal of Computational Chemistry*, 26(16), 1781–1802.
- Rash, J. E. (2010). Molecular disruptions of the pial syncytium block potassium siphoning and axonal saltatory conduction: Pertinence to neuromyelitis optica and other demyelinating diseases of the central nervous system. *Neuroscience*, 168(4), 982–1008.
- Robinson, R., Hampson, C. G. M., Munro, M. N., & Vaney, D. I. (1993). Unidirectional coupling of gap junctions between neuroglia. *Science (New York, N.Y.)*, 262(5136), 1072–1074.
- Scholl, U. I., Choi, M., Liu, T., Ramaekers, V. T., Hausler, M. G., Grimmer, J., ... Lifton, R. P. (2009). Seizures, sensorineural deafness, ataxia, mental retardation, and electrolyte imbalance (SeSAME syndrome) caused by mutations in KCNJ10. *Proceeding of the National Academy of Sciences United States of America*, 106(14), 5842–5847.
- Stensel, M. A., Spruijt, L., Burgt, I., Bladergroen, R. S., Vermeer, M., Steijlen, P. M., & Geel, M. (2005). A 2-bp deletion in the GJA1 gene is associated with oculo-dento-digital dysplasia with palmoplantar keratoderma. *American Journal of Medical Genetics Part A*, 132A(2), 171–174.
- Suchyna, T. M., Nitsche, J. M., Chilton, M., Harris, A. L., Veenstra, R. D., & Nicholson, B. J. (1999). Different ionic selectivities for connexins 26 and 32 produce rectifying gap junction channels. *Biophysical Journal*, 77(6), 2968–2987.
- Teubner, B., Odermatt, B., Guldénagel, M., Söhl, G., Degen, J., Bukauskas, F. F., ... Willecke, K. (2001). Functional expression of the new gap junction gene connexin47 transcribed in mouse brain and spinal cord neurons. *The Journal of Neuroscience*, 21(4), 1117–1126.
- Tress, O., Maglione, M., May, D., Pivneva, T., Richter, N., Seyfarth, J., ... Willecke, K. (2012). Pial gap junctional communication is essential for maintenance of myelin in the CNS. *The Journal of Neuroscience*, 32(22), 7499–7518.
- Trexler, E. B., Bukauskas, F. F., Kronengold, J., Bargiello, T. A., & Verselis, V. K. (2000). The first extracellular loop domain is a major determinant of charge selectivity in connexin46 channels. *Biophysical Journal*, 79(6), 3036–3051.
- Uhlenberg, B., Schuelke, M., Rüschenhoff, F., Ruf, N., Kaindl, A. M., Henneke, M., ... Gärtner, J. (2004). Mutations in the gene encoding gap junction protein $\alpha 12$ (Connexin 46.6) cause Pelizaeus-Merzbacher-like disease. *American Journal of Human Genetics*, 75(2), 251–260.
- Wang, H. Z., & Veenstra, R. D. (1997). Monovalent ion selectivity sequences of the rat connexin43 gap junction channel. *The Journal of General Physiology*, 109(4), 491–507.
- Wasseff, S. K., & Scherer, S. S. (2011). Cx32 and Cx47 mediate oligodendrocyte:astrocyte and oligodendrocyte:oligodendrocyte gap junction coupling. *Neurobiology of Disease*, 42(3), 506–513.
- Weber, P. A., Chang, H.-C., Spaeth, K. E., Nitsche, J. M., & Nicholson, B. J. (2004). The permeability of gap junction channels to probes of different size is independent on connexin composition and permeant-pore affinities. *Biophysical Journal*, 87(2), 958–973.
- Wingerchuk, D. M., Lennon, V. A., Lucchinetti, C. F., Pittock, S. J., & Weinshenker, B. G. (2007). The spectrum of neuromyelitis optica. *The Lancet Neurology*, 6(9), 805–815.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article. Additional Supporting Information may be found online in the supporting information tab for this article.

How to cite this article: Fasciani I, Pluta P, González-Nieto D, et al. Directional coupling of oligodendrocyte connexin-47 and astrocyte connexin-43 gap junctions. *Glia*. 2018;66: 2340–2352. <https://doi.org/10.1002/glia.23471>